

WHERE A NEURAL AUDIO CODEC PULLS PITCH TOWARD EQUAL TEMPERAMENT

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ABSTRACT

Neural vocoders pull tuning toward 12-tone equal temperament (12-TET). With tone pairs a fixed interval apart we locate the pull inside a neural audio codec: at low bitrate the decoder transmits lower partials and invents the rest above a band edge. The fundamental stays put; invented partials land 0.72 to 0.90 of the way onto the nearest 12-TET pitch’s ladder, and a full-band harmonic fit reports a displacement no partial underwent, 4.37 cents over 24 stimuli, 12.55 on a 440 Hz probe. Fine-tuning only EnCodec’s decoder on a corpus retuned by 33 cents moves its grid by 32.7 cents; classical codecs do not pull; beyond EnCodec and WavTokenizer the account does not yet predict which codecs do. Under it a round trip flips 22 of 200 Carnatic and 11 of 60 makam recordings toward 12-TET; a period tracker is spared, and fitting only below the edge removes the added error.

Index Terms— Neural audio codec, pitch, equal temperament, bandwidth extension, music information retrieval

1. INTRODUCTION

Modern audio models, from music generation [1] to spoken dialogue [2], operate on residual-vector-quantiser tokens from a neural audio codec [3, 4] and hear them through its decoder. Codec quality is reported as corpus-averaged scores [5], which cannot say whether a pitch displacement is grid-directed or which partials moved. Neural vocoders are known to pull output tuning toward 12-TET, the twelve-tone equal-tempered grid of 100-cent steps at A4 = 440 Hz, a cent being a hundredth of a semitone [6]; where inside a codec that happens, and which partials carry it, has not been measured.

We treat the codec as a measurement instrument. A pair of harmonic tones a controlled interval apart is coded, a blind estimator reads the decoded interval, and the residual is studied as the interval sweeps the octave and the reference pitch is detuned; the detuning is a *registration test* that separates a residual anchored to absolute pitch from one tied to the interval or the analysis.

Our contributions are:

- **Which partials move.** The fundamental and the partials below the edge stay put; those above it land 0.72 to 0.90 of the way onto the grid harmonic at 440 Hz in each of the seven measurable conditions that resolve an edge.
- **The ladder follows the corpus.** Fine-tuning only EnCodec’s decoder (probe tokens bit-identical across arms) on a corpus retuned by +33 cents moves the residual’s zero crossing by 32.7 ± 2.8 cents, a prediction logged before the arm ran. Flattening the corpus lowers the ladder fraction from 0.86 to 0.59; grid-preserving resamplings do not.
- **A measurement hazard, and its fix.** Under a full-band harmonic fit, one round trip flips 22 of 200 Carnatic and 11 of 60 Turkish makam recordings to fitting 12-TET better than their own measured degrees, every flip toward 12-TET; a period tracker is largely spared, and fitting no partial above the edge removes the added error, most of which is generic codec damage.

Relation to prior work. Two things were known, and neither locates the effect. A decoder may synthesise the spectrum above a band edge rather than transmit it: spectral band replication in HE-AAC [7], the folding in Opus’s CELT layer [8], and regeneration designed on purpose [9]; copy-up carries no frequency prior. And neural synthesis pulls tuning toward 12-TET: Berendes et al. [6] find mel-to-audio vocoders cluster retuned recordings near A4 = 440 Hz, at up to 18.7 cents, and fine-tune BigVGAN-V2 to preserve tuning [10]. Our claim is narrower and located: we measure which partials move and how far, describe the effect by an edge that rises with bitrate and a ladder the decoder’s corpus shifts, retune a corpus, which *translates* the grid, and follow the consequence for non-Western tuning, where the codec plays the tuning shift that confounds classifiers in [11]. A full-band estimator forming one pitch across partials is the algorithmic analogue of virtual pitch [12]. Tuning analysis on these corpora rests on pitch histograms [13] from salience-based melody extraction [14]; such tools inherit the hazard in proportion to the weight they place above the edge, which our full-band fit weights more than a salience sum.

2. MEASURING THE CODEC TRANSFER FUNCTION

2.1. Transfer function and grid bias

A neural audio codec comprises an encoder, a residual vector quantiser and a decoder, all learned from a training corpus. Let $x(\theta)$ be a stimulus in which one property takes the value θ , here a musical interval in cents, and $\mathcal{E}$ an estimator that recovers that property from a waveform. The *transfer function* of a codec C along θ and its residual are

$$T_C(\theta) = \text{med}_{\text{trials}} \mathcal{E}[C(x(\theta))], \quad r_C(\theta) = T_C(\theta) - \theta, \quad (1)$$

medians over the random phases and gains of repeated trials. Let $g(\theta) = 100 \cdot \text{round}(\theta/100)$ be the nearest 12-TET interval and $\delta(\theta) = \theta - g(\theta) \in [-50, 50]$ the signed distance to it. *Grid bias* is the residual’s component directed toward the grid,

$$b(\theta) = -\text{sign}(\delta(\theta)) r_C(\theta), \quad (2)$$

Every verdict uses one statistic, the *off-grid bias* $\tilde{b}$: the median of b over trials with $|\delta| \geq 30$, read at the on-grid reference. Intervals are 99.55% percentile bootstraps over one condition’s trials (Bonferroni over the eleven codec rows), covering trial scatter only.

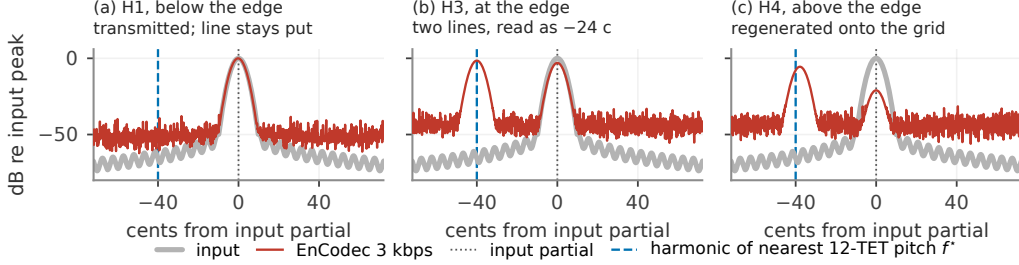


Fig. 1. What moves, and what does not. A 440 Hz complex detuned +40 cents, EnCodec at 3 kbps. H_k is partial k ; dashed, kf^* , with f^* the 12-TET pitch nearest f_0 ; zero marks the input partial. In (b) the coded trace is shifted up for display; -24 is the two lines’ centroid (Table 2).

2.2. Registration test and pull rule

Any quantiser or estimator can produce a 100-cent-periodic residual; a learned grid has its zero crossings at 12-TET *pitch*s, a statement about absolute frequency. Detuning the reference by d cents therefore separates two hypotheses: under absolute anchoring the residual profile translates by d , under interval anchoring it does not. Regressing the fitted sinusoid’s phase on d ,

$$\varphi(d) = \varphi_0 + \sigma (2\pi/100) d, \quad \sigma = 1 \text{ (pitch), } 0 \text{ (interval),} \quad (3)$$

makes the slope σ the test. A condition *registers* when the 95% t interval of σ over the ten detunings (unadjusted) lies inside 1 ± 0.15 , a margin fixed in advance; this is an equivalence test [15], not a failure to reject. Registration is necessary but not sufficient. We claim a learned grid only where a condition also meets the pre-specified *pull* rule: on-grid phase within 45° of 180° and an off-grid bias interval that clears zero. On our data the bias interval decides every verdict. Two pre-specified guards refuse a registration fit when the fitted amplitude’s coefficient of variation across detunings exceeds 0.35 (DAC 44k, 0.41) or fewer than 25% of trials survive the tracking gate (SNAC 24k, 7%).

2.3. Band edge and ladder fraction

A harmonic-fit estimator combines the first eight partials into one number, so its residual does not say which partials moved. We therefore read the decoded spectrum directly. Let $\hat{f}_k$ be the power-weighted centroid of the lines within 100 cents of the k th input partial kf_0 and within 6 dB of the strongest among them, f^* the 12-TET pitch nearest f_0 , and $\delta_0 = 1200 \log_2(f_0/f^*)$ the tone’s own detuning. The per-partial displacement and its *ladder fraction* are

$$s_k = 1200 \log_2(\hat{f}_k/kf_0), \quad \ell_k = -s_k/\delta_0, \quad (4)$$

so $\ell_k = 0$ for a partial left on the input’s harmonic ladder and $\ell_k = 1$ for one moved onto the grid pitch’s ladder; a ratio, so an overshooting partial reads above 1. Partial k counts as *transmitted* when its decoded line is within 6 dB of the input’s, and as *regenerated* when it has fallen more than 10 dB below that and the strongest line within 100 cents sits elsewhere. The *band edge* f_e is the lowest regenerated partial, right-censored at $0.95f_s/2$ when there is none, and is reported per condition as a mean over the registers where it resolves. The ladder fraction $\bar{\ell}$ is the mean ℓ_k from the edge up, read at $|\delta_0| \geq 20$, where it is flat across 5 to 50 cents.

2.4. Stimuli, estimator, codecs and pre-registration

Each stimulus is a pair of harmonic complex tones in sequence, the first at a reference f_0 and the second at $f_0 2^{g/1200}$, with $1/n$ partial amplitudes and phases redrawn per repetition; the pitch probe uses eight partials over 500 ms, the spectral sweep twelve over one second. The interval sweeps 0 to 1200 cents in 5-cent steps at twenty draws and the reference over eleven values spanning a semitone in 10-cent steps (the 100-cent value folds onto 0, giving ten detunings), $241 \times 11 \times 20 = 53,020$ trials, repeated at 220 and 880 Hz, over four octaves, six spectral envelopes by four registers and a source-filter vowel. F_0 is estimated in three stages over the whole tone: YIN [16] and harmonic summation each give a blind coarse estimate, and a harmonic least-squares refinement searches a ± 60 -cent window around each expected partial, locates each line by parabolic interpolation of the log-magnitude spectrum and combines the first eight with weights $w_k \propto k^2 |X_k|^2$. The YIN-seeded value is reported; the two seeds’ disagreement is the *cross-check*, a robustness gate at 5 cents on sweeps and 20 on recordings. A learned tracker [17] is excluded because, trained on music, it would carry the prior we measure; pYIN [18] returns as a control in Section 3.4. The only exclusion is a tracking gate at 200 cents from nominal, which removes 12 of the 53,020 trials of the EnCodec 3 kbps headline run. A *band-limited* reading is the same estimator with the partial set truncated to those below the codec’s measured edge, 1.32 kHz for EnCodec at 3 kbps, so it needs no uncoded original.

We evaluate EnCodec [3], DAC [4], Mimi [2], SpeechTokenizer [19], SNAC [20] and WavTokenizer [21]; an operating point is a bitrate or a count Q of quantiser stages. BigVGAN [22], a neural vocoder, and Opus [8], MP3 and HE-AAC with SBR serve as controls. The pre-registration placed the residual in the quantiser, a codebook learned from 12-TET music rounding an off-grid pitch toward an on-grid one, and predicted a sawtooth residual in δ (P1), a positive off-grid bias (P2), a bias falling with bits per latent dimension (P3) and registration to absolute pitch (P4); it also fixed the pull rule, the registration window and the tracking gate. Registration and (P2) hold; (P1) fails (Section 3.2); (P3) is out of regime, EnCodec’s 0.16 to 2.5 bits per latent dimension lying outside the derivation’s assumptions. The pre-registration document and its amendment log are released with the code. All else below is exploratory and unadjudged except the registered pass (Section 3.3).

Table 1. Which conditions pull, and where their partials land. 440 Hz reference. σ : registration slope $\pm$ half-width of its 95% t interval; registers when inside 1 ± 0.15 . Edge in kHz; dash, censored. $\bar{\ell}$: mean over the two detuning signs, DAC 16k’s at 880 Hz. Bias: $\bar{b}$, 99.55% trial bootstraps over the eleven codec rows (Bonferroni), 95% for the controls; Opus’s median sits at zero, so its interval is one-sided; MP3’s 0.01-cent residual carries no phase.

Condition	σ	$\pm$	Edge	$\bar{\ell}$	Bias (c)
<i>Pulls (registers, bias clears zero)</i>					
EnCodec 24k, 3 kbps	1.001	.009	1.3	0.90	12.55 [10.71, 13.67]
EnCodec 24k, 24 kbps	0.997	.005	2.2	0.72	8.03 [6.45, 8.81]
WavTok., 0.9 kbps	1.004	.007	0.9	0.84	7.61 [6.44, 8.52]
EnCodec 48k, 6 kbps	1.019	.015	1.6	0.79	6.54 [5.83, 7.80]
SNAC 32k	0.964	.038	3.1	0.86	0.61 [0.18, 1.02]
EnCodec 24k, vowels	1.011	.021	–	–	0.43 [0.19, 0.60]
<i>No pull</i>					
Mimi, Q8	0.958	.028	3.4	0.75	1.19 [−0.15, 2.86]
DAC 16k, Q6	1.020	.063	4.4	0.85	−0.13 [−0.55, 0.07]
SpeechTok., vowels	0.856 [†]	.032	–	–	−0.03 [−0.21, 0.14]
DAC 24k, Q8	0.979	.028	–	–	0.01 [−0.22, 0.13]
SNAC 44k	0.995	.041	–	–	−0.14 [−0.29, 0.07]
<i>Refused by the guards: spectral read only</i>					
DAC 44k, Q4 [‡]	–	–	res.	0.14	–
SNAC 24k	–	–	res.	0.28	–
<i>Controls</i>					
BigVGAN (vocoder)	0.997	.011	2.6	0.68	4.82 [3.91, 5.64]
HE-AAC SBR, 32 kbps	–	–	1.3*	0.04	0.05 [−0.31, 0.42]
Opus, 6 kbps	0.929	.137	–	–	0.00 [0.00, 0.42]
MP3, 16 kbps	n/a	–	–	–	0.00 [0.00, 0.00]

[†]Outside the window; registers under the cross-check (0.989 $\pm$.085). *SBR crossover set to EnCodec’s 1.32 kHz edge. [‡]Ladder read at Q8. res.: resolved.

3. RESULTS

3.1. Six conditions pull, four by over five cents

The residual is anchored to absolute pitch. For EnCodec at 3 kbps the registration slope is 1.0010 [0.9922, 1.0099] at $R^2 = 0.99988$, inside the pre-specified window. Over a population of twenty-four stimuli, six spectral envelopes at four registers, the median off-grid bias is $\bar{b} = 4.37$ cents [1.73, 9.13], a bootstrap over envelopes and registers read as spread rather than coverage. Conditions are compared on one tone, a 440 Hz complex that reads 12.55 cents through EnCodec and sits at that population’s 92nd percentile. The same tone reads 13.72 as fitted sinusoid amplitude, which overshoots a saturating profile.

Key findings (Table 1). (1) Eleven of the thirteen neural conditions are measurable; ten of them register, and the eleventh, SpeechTokenizer, only under the cross-check. (2) Six pull under the pre-specified rule: EnCodec at 3, 24 and 6 kbps and WavTokenizer by more than five cents, SNAC 32k and EnCodec on vowels by under a cent. (3) The nulls differ in kind: DAC 16k pulls at 880 and 1760 Hz, by 7.0 to 7.6 and 10.9 to 11.3 cents, where partials clear its 4.4 kHz edge; three censored nulls have no resolvable edge; Mimi’s interval covers zero and its +1.19 sits below its 2.0-cent 80%-power floor, a non-detection rather than a bounded null. Floors are per condition

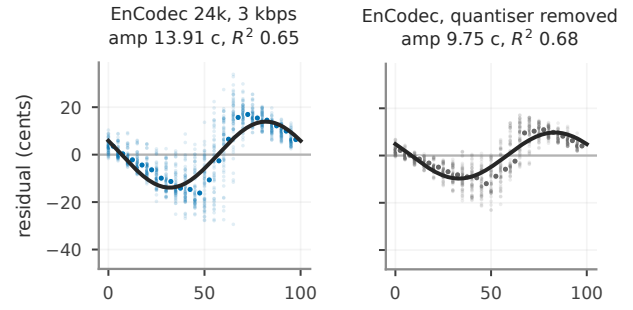


Fig. 2. The residual r_C over one semitone at 440 Hz. Left, EnCodec 24k at 3 kbps; right, EnCodec with the quantiser removed. Points, trials and binned medians; curve, fitted sinusoid (a single-condition fit; the text’s 13.72 is the mean over ten detunings); x axis, cents above the 12-TET pitch below.

Table 2. Which partials move. Displacement of each decoded partial, cents, EnCodec 24k at 3 kbps, 440 Hz, $\delta_0 = \pm 40$, on the spectral-sweep stimulus (twelve partials, one second); means over repetitions, whereas Table 1’s $\bar{\ell}$ averages $|\delta_0| \geq 20$. H3, at the edge, splits across two lines and reads at their centroid; *Est.*, the eight-partial estimator’s reading on that stimulus.

δ_0	H1	H2	H3	H4	H5	H6	H7	H8	Est.
−40	0.0	0.0	26.9	40.2	51.9	42.1	35.1	44.6	20.7
+40	−0.2	0.0	−24.0	−37.9	−27.5	−35.1	−42.8	−37.5	−16.9

(0.52 to 2.00 cents); verdicts follow the interval rule, under which SNAC 32k and the vowel condition clear zero near their floors of 0.71 and 0.52. (4) The classical controls do not pull.

3.2. The fundamental stays put; the invented partials move

For EnCodec the line at f_0 sits within 0.2 cents of the input at each of three bitrates and every partial below the edge within 2 cents; above it, at 3 kbps, partials land 0.95 and 0.90 of the way to the grid harmonic at 220 and 440 Hz, single partials overshooting (Table 2). BigVGAN, a vocoder, also moves the fundamental (3.1 cents), which a recording-level estimate [6] cannot separate from this.

The edge rises with bitrate, and the bias falls with it. For EnCodec 24 kHz the edge moves from the second partial at 1.5 kbps (spectral sweep) to the fifth at 24 kbps, about 0.9 to 2.2 kHz at a 440 Hz reference, and the bias reads 12.55 cents at 3 kbps and 8.03 at 24 kbps. Against the quantiser account, (P1) fails: the residual saturates (Fig. 2); a clipped profile with a 22-cent knee beats the sawtooth and the sinusoid on AIC by 15.8 and 14.2. And running the encoder straight into the decoder, which removes the quantiser at a matched 2.2 kHz edge, leaves the all-trial median bias, the statistic the bypass was pre-specified on, at 5.66 [5.28, 6.06] against the 24 kbps condition’s 5.73 [5.36, 6.11], so the effect does not require a quantiser; nor do the tokens carry a grid: the 2341 code-assignment boundaries located within the semitone show no concentration (Rayleigh $p = 0.86$).

3.3. The grid follows the decoder’s training corpus

Every EnCodec arm fine-tunes the same EnCodec decoder for 4000 steps with encoder and quantiser frozen (AdamW at 10^{-5} , batches of sixteen 0.5-s segments, multi-resolution STFT loss plus ten times the waveform ℓ_1 , no adversarial term) and is read at 3 kbps. Fine-tuning itself raises the edge from 1.3 to 2.0 kHz and removes most of the amplitude whatever the corpus (13.72 cents stock, 4.51 on unmodified GTZAN [23]), so the corpus contrasts are read within that shared edge. The arms differ only in the tuning of the GTZAN clips: unmodified; resampled by one semitone or by 0 or 100 cents per clip (grid-preserving; the latter *magnitude-matched* to the flattening); *flattened*, each clip shifted by an offset drawn uniformly on $[0, 100]$ cents, which removes the grid; and resampled by +33 cents. Against 12-TET the ladder fraction reads 0.89 unmodified, 0.88 and 0.86 for the two grid-preserving controls, 0.59 flattened and 0.11 returned; the retuned arm reads 0.87 against its own grid.

Retuning moves the grid by about the predicted amount.

The decoder fine-tuned on unmodified clips has its *attractor*, the fitted profile’s zero crossing, at 6.9 [5.4, 8.3] cents, part of it arithmetic since kf^* is not the chromatic pitch at $k = 5, 7$ (a +1.6-cent gap); the grid-preserving controls sit at 7.1 and 6.6. Resampling the corpus uniformly upward by 33 cents moves it to 39.6 [37.1, 42.0] against 6.9 + 33 predicted, a displacement of 32.7 ± 2.8 cents (mean $\pm$ sd over five seeds); the same manipulation moves Mimi’s crossing from 5.7 to 38.4 [34.2, 42.6] and WavTokenizer’s from 4.9 to 37.9 [34.6, 41.2]. Flattening lowers the ladder fraction to 0.59 against 0.86 for the magnitude-matched control, the amplitude to 3.75 against 4.69, and lowers $\bar{b}$ by 0.87 cents [0.74, 1.00] more than that control does over ten draws of the offsets. The remaining 0.59 reflects the pre-trained start: a bandwidth extender trained from scratch to fill the band above EnCodec’s 1.32 kHz edge (4.1M parameters, five seeds) reads 0.68 [0.55, 0.81] on GTZAN as recorded, 0.14 [0.00, 0.28] on the same audio flattened. A decoder fine-tuned on Saraga [24] reads 0.81 against that corpus’s own histogram of scale degrees and 0.22 against 12-TET.

Held-out and registered passes. We froze a three-tier rule on $\bar{\ell}$ (*ladder* ≥ 0.70 , *part* 0.20 to 0.70, *input*’s below 0.20) and scored twenty-six held-out checkpoints, predicting the tier from architecture and bitrate: on the twenty from untouched families the rule is right on 12 against 10 for always predicting *part* (exact McNemar $p = 0.815$). We then changed the predictor to how many of the probe’s eight partials clear the checkpoint’s own measured edge, which tests the edge-to-ladder link rather than prediction from architecture, registered it with that comparator and sample size, and scored 15 fresh checkpoints: right on 11 against 5 ($p = 0.070$), not distinguishable from always predicting *ladder*, that sample’s majority tier ($p = 0.375$). The account’s reach beyond the primary set is not established.

3.4. Cost on real recordings, and a remedy

Protocol. Scale degrees are the modes of each recording’s within-octave F_0 histogram, folded so that 12-TET sits at $A4 = 440$ Hz, matched by nearest neighbour to the corpus’s aggregate measured positions placed at the recording’s annotated tonic, and scored as the median absolute deviation from them; the reference is built from uncoded audio with the same pipeline, leave-one-recording-out. The pipeline reads 64-ms frames with the cross-check at 20 cents, which retains 23 to 40% of frames.

Size and direction. Frame by frame, the off-grid bias under EnCodec at 3 kbps is 2.49 cents [2.05, 2.98] on tuning-flattened

Table 3. What one round trip costs a tuning analysis. Median scale-degree error added by coding, in cents, against each corpus’s own measured degrees (uncoded: 4.1 on Saraga, 5.2 on makam); flips: recordings whose best-fitting system changes after coding, counted in either direction; every full-band EnCodec flip moved toward 12-TET. Matched Opus: the bitrate whose coded degree error comes closest to EnCodec’s from below (4.6 and 5.2 kbps).

Reading	Added error (c)	Flips
<i>Saraga, Carnatic, n = 200</i>		
full-band fit, EnCodec 3 kbps	3.2 [2.6, 3.8]	22 (11%)
full-band fit, matched Opus	2.8	2
band-limited fit, EnCodec	0.2	4
pYIN, EnCodec	0.7 [0.2, 1.2]	6 (3%)
<i>Turkish makam, n = 60</i>		
full-band fit, EnCodec 3 kbps	4.4 [3.5, 5.4]	11 (18%)
full-band fit, matched Opus	3.7	2
band-limited fit, EnCodec	0.2	2

GTZAN and 2.63 [2.17, 3.14] on Saraga, and, as the account predicts, 2.68 cents [1.38, 4.30] larger where partials clear the edge on Saraga, not under Opus. Table 3 gives the cost on Saraga (Carnatic) and Turkish makam [25] recordings: every EnCodec flip moves toward 12-TET, the discordant pairs against matched Opus split 22 to 2 on Saraga and 10 to 1 on makam (exact McNemar, $p = 4 \times 10^{-5}$ and 0.012), and the nineteen flips across all control arms (pYIN excluded) split ten toward 12-TET and nine away.

The remedy. Fitting only below the edge takes both corpora from 3.2 and 4.4 cents of added degree error to 0.2, both intervals covering zero. Most of what it removes is generic codec damage: the remainder against matched Opus is 0.4 and 0.7 cents, neither excluding zero. pYIN, a period tracker, is largely spared. The decoder tuned on Saraga nearly removes the Carnatic penalty, 0.3 [−0.2, 0.8] against 1.3 for a GTZAN-tuned decoder, while making makam, which it never saw, worse, 6.6 [5.5, 7.7] with flips toward Carnatic positions; a pull toward 12-TET would cost both.

4. CONCLUSION

Where a pulling codec resolves a band edge, the decoder invents the partials above it near the harmonics of the nearest 12-TET pitch, a full-band fit over them reports a displacement no partial underwent, and retuning the fine-tuning corpus by 33 cents moves the grid by 32.7 cents.

Limitations. The band-edge account is post hoc, and its registered pass reaches $p = 0.070$ without separating from the majority tier, so its reach beyond EnCodec and WavTokenizer is not established. Downstream, the grid-attributable degree error does not exclude zero; the grid’s cost is directional.

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6. COMPLIANCE WITH ETHICAL STANDARDS

This work uses synthetic stimuli and existing public corpora, GTZAN, Saraga and the Turkish makam corpus, each under its own licence; no human subjects were involved and no new data were collected, so no ethical approval was required. No corpus audio is redistributed.

7. REFERENCES

- [1] Jade Copet, Felix Kreuk, Itai Gat, Tal Remez, David Kant, Gabriel Synnaeve, et al., “Simple and controllable music generation,” in *Proc. NeurIPS*, 2023.
- [2] Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, et al., “Moshi: a speech-text foundation model for real-time dialogue,” *arXiv preprint arXiv:2410.00037*, 2024.
- [3] Alexandre Défossez, Jade Copet, Gabriel Synnaeve, and Yossi Adi, “High fidelity neural audio compression,” *Trans. Mach. Learn. Res.*, 2023.
- [4] Rithesh Kumar, Prem Seetharaman, Alejandro Luebs, Ishaan Kumar, and Kundan Kumar, “High-fidelity audio compression with improved RVQGAN,” in *Proc. NeurIPS*, 2023.
- [5] Haibin Wu, Ho-Lam Chung, Yi-Cheng Lin, Yuan-Kuei Wu, Xuanjun Chen, Yu-Chi Pai, et al., “Codec-SUPERB: An in-depth analysis of sound codec models,” in *Findings of ACL*, 2024, pp. 10330–10348.
- [6] Hans-Ulrich Berendes, Ben Maman, and Meinard Müller, “Tuning matters: Analyzing musical tuning bias in neural vocoders,” in *Proc. ISMIR*, 2025, pp. 166–173.
- [7] Martin Dietz, Lars Liljeryd, Kristofer Kjörling, and Oliver Kunz, “Spectral band replication, a novel approach in audio coding,” in *AES 112th Conv.*, 2002, Convention Paper 5553.
- [8] Jean-Marc Valin, Koen Vos, and Timothy B. Terriberry, “Definition of the Opus audio codec,” RFC 6716, Internet Engineering Task Force, 2012.
- [9] Woongjib Choi, Byeong Hyeon Kim, Hyungseob Lim, Inseon Jang, and Hong-Goo Kang, “Neural spectral band generation for audio coding,” in *Proc. Interspeech*, 2025.
- [10] Hans-Ulrich Berendes, Ben Maman, and Meinard Müller, “Fine-tuning BigVGAN-V2 for robust musical tuning preservation,” in *Proc. IEEE ICASSP*, 2026.
- [11] Yi Qin and Alexander Lerch, “Tuning frequency dependency in music classification,” in *Proc. IEEE ICASSP*, 2019, pp. 401–405.
- [12] Ernst Terhardt, Gerhard Stoll, and Manfred Seewann, “Algorithm for extraction of pitch and pitch salience from complex tonal signals,” *J. Acoust. Soc. Am.*, vol. 71, no. 3, pp. 679–688, 1982.
- [13] Barış Bozkurt, Ruhi Ayangil, and André Holzapfel, “Computational analysis of Turkish makam music: Review of state-of-the-art and challenges,” *J. New Music Res.*, vol. 43, no. 1, pp. 3–23, 2014.
- [14] Justin Salamon and Emilia Gómez, “Melody extraction from polyphonic music signals using pitch contour characteristics,” *IEEE Trans. Audio, Speech, Lang. Process.*, vol. 20, no. 6, pp. 1759–1770, 2012.
- [15] Donald J. Schuirmann, “A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability,” *J. Pharmacokinet. Biopharm.*, vol. 15, no. 6, pp. 657–680, 1987.
- [16] Alain de Cheveigné and Hideki Kawahara, “YIN, a fundamental frequency estimator for speech and music,” *J. Acoust. Soc. Am.*, vol. 111, no. 4, pp. 1917–1930, 2002.
- [17] Jong Wook Kim, Justin Salamon, Peter Li, and Juan Pablo Bello, “CREPE: A convolutional representation for pitch estimation,” in *Proc. IEEE ICASSP*, 2018, pp. 161–165.
- [18] Matthias Mauch and Simon Dixon, “pYIN: A fundamental frequency estimator using probabilistic threshold distributions,” in *Proc. IEEE ICASSP*, 2014, pp. 659–663.
- [19] Xin Zhang, Dong Zhang, Shimin Li, Yaqian Zhou, and Xipeng Qiu, “SpeechTokenizer: Unified speech tokenizer for speech large language models,” in *Proc. ICLR*, 2024.
- [20] Hubert Siuzdak, Florian Grötschla, and Luca A. Lenzendörfer, “SNAC: Multi-scale neural audio codec,” in *NeurIPS Workshop on Audio Imagination*, 2024.
- [21] Shengpeng Ji, Ziyue Jiang, Wen Wang, Yifu Chen, Minghui Fang, Jialong Zuo, et al., “WavTokenizer: An efficient acoustic discrete codec tokenizer for audio language modeling,” in *Proc. ICLR*, 2025.
- [22] Sang-gil Lee, Wei Ping, Boris Ginsburg, Bryan Catanzaro, and Sungroh Yoon, “BigVGAN: A universal neural vocoder with large-scale training,” in *Proc. ICLR*, 2023.
- [23] George Tzanetakis and Perry Cook, “Musical genre classification of audio signals,” *IEEE Trans. Speech Audio Process.*, vol. 10, no. 5, pp. 293–302, 2002.
- [24] Ajay Srinivasamurthy, Sankalp Gulati, Rafael Caro Repetto, and Xavier Serra, “Saraga: Open datasets for research on Indian art music,” *Empirical Musicology Review*, vol. 16, no. 1, pp. 85–98, 2021.
- [25] Burak Uyar, Hasan Sercan Atlı, Sertan Şentürk, Barış Bozkurt, and Xavier Serra, “A corpus for computational research of Turkish makam music,” in *Proc. DLFM*, 2014, pp. 1–7.